INSTRUCTIONS

PRELIMINARY GEK-24962

DIVIDER, 193X542A_G01–G04

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

PRELIMINARY INSTRUCTIONS DIVIDER, 193X542A_G01–G04

1.0 GENERAL

This instruction provides basic information regarding the subject card. Refer to the system elementary diagrams for information relating to the overall system operation.

2.0 DESCRIPTION

The card contains either one or two integrated circuit divider chips with the following group arrangements:

G01 and G03 one channel
G02 and G04 two channels

The G01 and G03 card, and the G02 and G04 card are both mechanically and functionally interchangeable. Different groups for the equivalent functions identify use of different electronic devices. A transistor power supply is furnished to convert from ± 20V to ± 15V.

PERFORMANCE

The card will have the following characteristics while exposed to the conditions indicated:

2.11 Division
With +20V and COM applied to tabs 31, 2 and 15 respectively, the card will generate a nominal output voltage of

$$OP = 10 \cdot N/D$$

where the input signal voltages N and D are within the ranges of

$.1V \leq D \leq 10V$ and
$|N| \leq D$

2.12 Overall Accuracy
Total error (25°C): ± .65% max. or ±0.65V for $D > .1V$
Total error vs temp: ± .033%/°C or 3.3mV/°C
Total error vs. power supply: ± .15%/% (±0.15% at 25°C and 0.033%/°C of error is caused by power supply variation)
Warm-up time to rated performance: 5 min.

2.13 Output
Max. output voltage: ± 10V
Max. output current: ± 5mA
Output Impedance: .1 ohm

2.14 Input
Rated numerator voltage, N: ± 10V with $|N| \leq D$
Rated denominator voltage, D: ± 10V with $N > 0$
Max. safe input voltage, N, D: ± 15V
Input Impedance: N–9K ohm, D–25K ohm

2.15 Power Supply
Range: 15V ± .15V at tab 26, –15V ± .15V at tab 8
Quiescent current: +15mA, –9mA at ±15V, +31mA, –25mA at ±20V.

2.16 Temperature
Rated performance: 0°C to +70°C
Max. operating: –25°C to +85°C

3.0 ADJUSTMENTS

There are no adjustments on this card.

4.0 TROUBLESHOOTING

4.1 Check for +15 ±1 volts at tab 26, –15 ±1 volts at tab 8

4.2 With +20 volts, –20 volts and Com applied to tabs 31, 2 and 15 respectively, apply 10 volts to N1 and D1 or N2 and D2. Card output for this condition should be 10 volts ± 0.65%.

If the output does not fall within this value, the card should be replaced.

4.3 Use an oscilloscope to determine if excessive noise on any of the input signals causes a distortion of the output signal during system operation. Filtering of the input signal(s) may be necessary to improve operation.

GENERAL ELECTRIC

36B590244AB

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UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

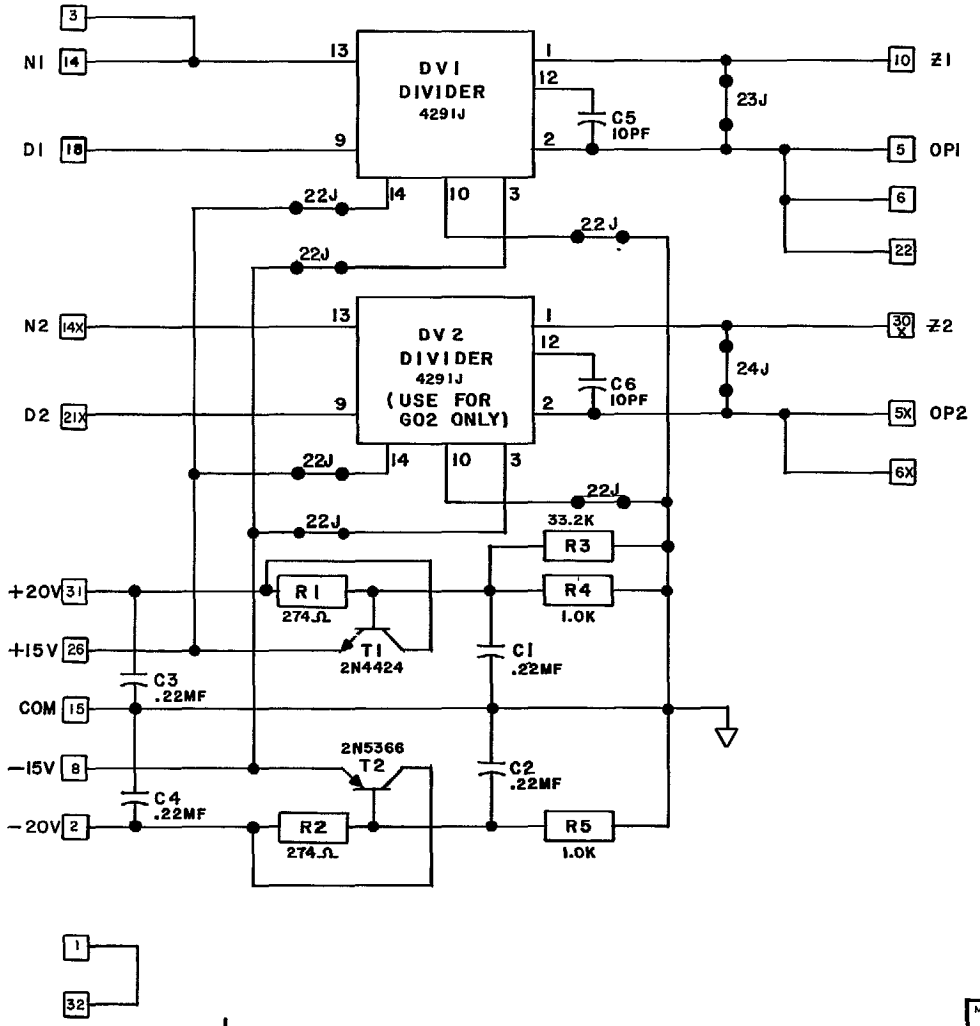
REV NO. 0

CONT ON SHEET | SH NO

TITLE ELEMENTARY DIAGRAM
DIVIDER CARD
BURR-BROWN, 4291J

FIRST MADE FOR STANDARD LINE

193X542ABG01, G02



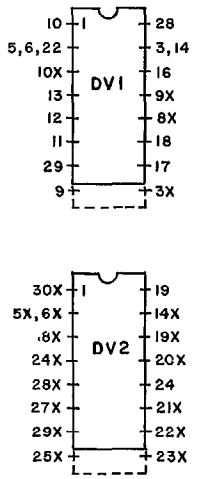
NOTES:

1. DIVIDER

$$OP = \frac{10 \cdot N}{D}$$

$$0.1V < D < 10V$$

IC CHIP POINTS RUN DIRECTLY TO TABS BUT ARE NOT SHOWN ON ELEM.



REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY J.R. Baker OCT. 11, 1977

ISSUED Carl Mork FEB 16, 1978

APPROVALS

SPEED VARIATOR DIV OR DEPT

ERIE, PA. LOCATION

36B590244AB

CONT ON SHEET | SH NO

FF 806 F(1 77)

GENERAL ELECTRIC

36C764179AB

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
DIVIDER CARD
FIRST MADE FOR STANDARD LINE
193X542ABG01, G02

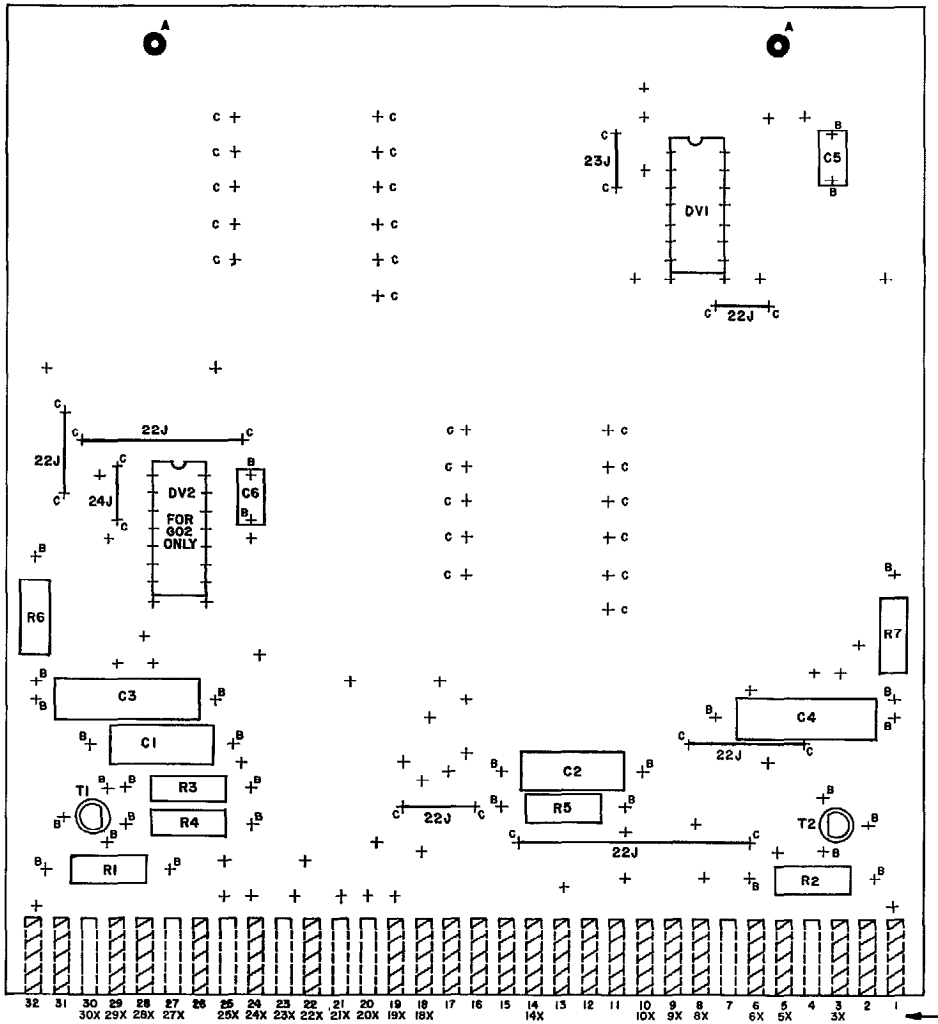
HOLE TABULATION

ALL HOLES 032 DIA EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QUAN
A	157	2
B	.040	32
C	.052	38

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.016} \times 5.130^{+.002}_{-.008}$
4. THIS CARD HAS GOLD PLATED TABS ON BOTH SIDES. TABS 1 THRU 32 ARE LOCATED ON THE REVERSE SIDE. TABS 1X THRU 32X ARE LOCATED ON THE COMPONENT SIDE OF THE CARD. TAB 1X IS OPPOSITE TAB 1 AND ETC. THE TAB NUMBERS SHOWN ARE THOSE USED ON THIS CARD.

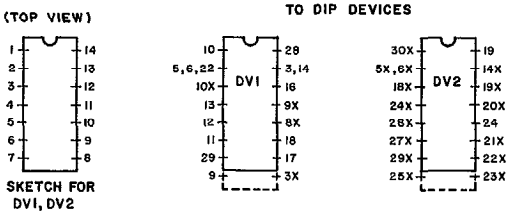


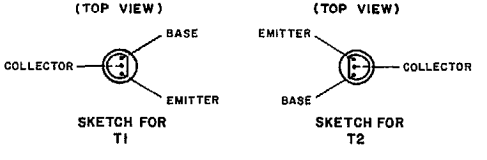
SEE NOTE 1 & 2

SEE NOTE 4

GROUP	KEY LOCATIONS			
G01	8-9	12-13	26-27	
G02	8-9	12-13	19-20	26-27

TAB CONNECTIONS TO DIP DEVICES





2X	36A358055AC005	36A358056AC005	1	—
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

REV NO 0

MADE BY G.R.Baker OCT. 24,1977
ISSUED Earl Mrak FEB. 16, 1978
APPROVAL
speed variator
ERIE, PA., USA

36C764179AB

FOLD

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO O

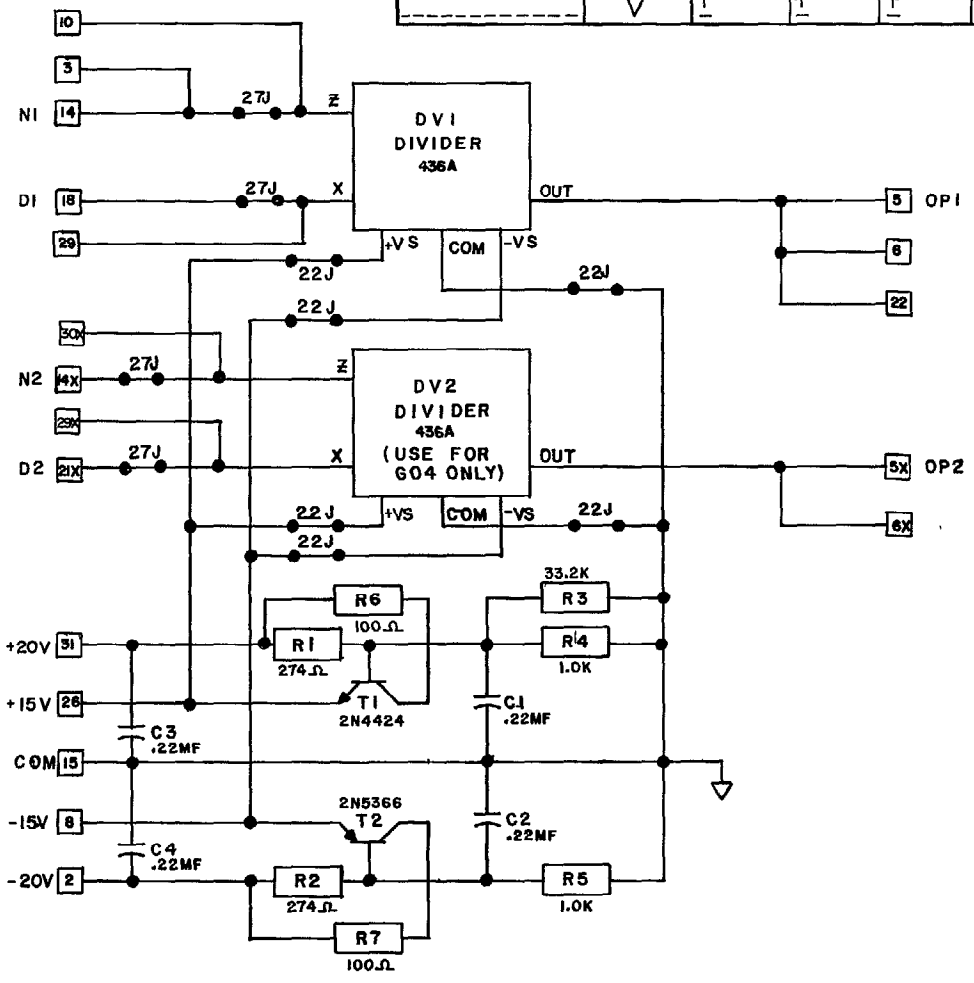
CONT ON SHEET SH NO

GENERAL ELECTRIC

36B590246AB
CONT ON SHEET SH NO

TITLE ELEMENTARY DIAGRAM
DIVIDER CARD
ANALOG DEVICES, 436A

FIRST MADE FOR STANDARD LINE
193X542ABG03 & G04



NOTES

1. DIVIDER

$$OP = \frac{10 \cdot N}{D}$$

$0.1V < D < 10V$

REVISIONS

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PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY J.R. Baker OCT. 10, 1977
ISSUED Earl Mook FEB. 16, 1978

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

36B590246AB
CONT ON SHEET SH NO.

FF 806-F(1 77)

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

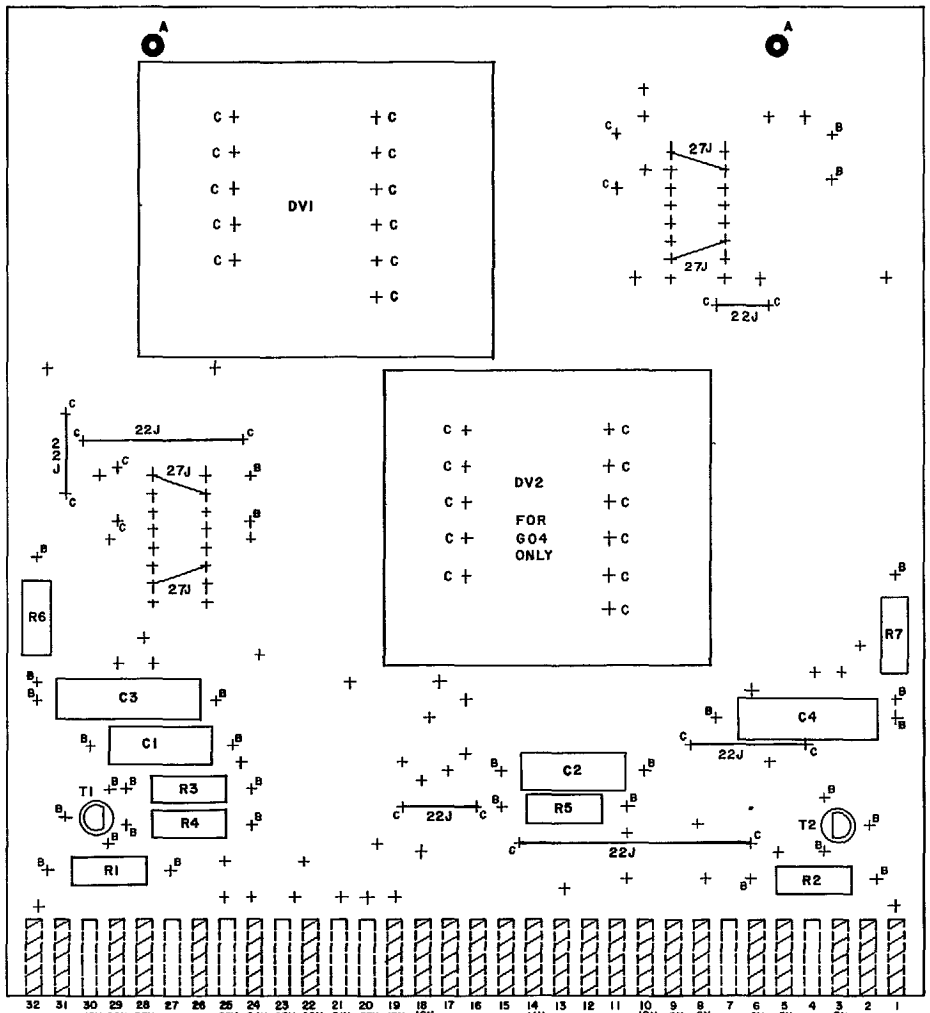
TITLE PRINTED CIRCUIT DIAGRAM
DIVIDER CARD
FIRST MADE FOR STANDARD LINE
193X542AB G03, G04

36C764182AB
CONT ON SHEET
SH NO

HOLE TABULATION

ALL HOLES 032 DIA EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA	QUAN
A	157	2
B	.040	32
C	.052	38



GROUP	KEY LOCATIONS			
G03	8-9	12-13	26-27	
G04	8-9	12-13	19-20	26-27

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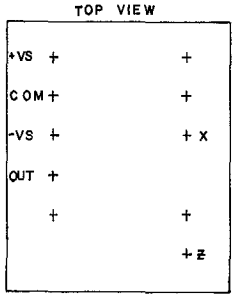
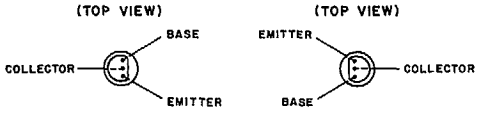


DV1 , DV2



SKETCH FOR T1 — SKETCH FOR T2

2X	36A358055AC005	36A358056AC005	1	—
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	FRONT SPACE UNIT REQUIREMENT	BACK

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5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY J.R. Baker OCT 21, 1977
ISSUED Paul Novak FEB. 16, 1978
APPROVALS
SPEED VARIATOR ERIE, PA.
DIV. OR DEPT. LOCATION
36C764182AB
CONT ON SHEET SH NO

REV NO 0

SPEED VARIATOR PRODUCTS OPERATION
GENERAL ELECTRIC COMPANY – DIRECT CURRENT MOTOR & GENERATOR DEPARTMENT
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC

GEK-24962 (8-78) 1.5M (F)